

Generating row–column field experimental designs with good neighbour balance and even distribution of treatment replications

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Abstract

Blocking is a standard strategy for improving the precision of randomized experimental designs. In field trials, it is usually beneficial to impose blocks in both row and column directions. A common problem with classical randomized row–column designs is that replications of a treatment may be clustered and that some pairs of treatments appear next to each other rather more often than others. Such patterns adversely affect certain design properties which we refer to as neighbour balance (NB) and evenness of distribution (ED), and they may be particularly harmful if some of the treatments are very susceptible to environmental stresses. Here, we consider an experimental design strategy for blocked experiments that achieves good NB&ED properties. The approach is exemplified using five examples, spanning non-resolvable and resolvable replicated designs as well as partially replicated designs.

KEYWORDS

A-efficiency, average efficiency factor, experimental design, latinization, row–column design

1 | INTRODUCTION

Randomized field or greenhouse experiments often show spatial trends and heterogeneity across the layout due to environmental factors. Spatial trends and heterogeneity tend to be particularly pronounced in experiments conducted under various forms of induced or natural biotic and abiotic stress, thus causing substantial error and adversely affecting the precision of treatment comparisons (Braun et al., 1992; Cooper et al., 1997; Hübner et al., 2012; Kumar et al., 2007; Qiao et al., 2000; Singh et al., 2003; Weber et al., 2012). Blocking may be used to control for such trends and heterogeneity, thereby improving precision compared with complete randomization, and many blocked designs have been proposed for the purpose over the last century (Edmondson, 2005; Verdooren, 2020). The simplest blocked design is the randomized complete block design. When the number of treatments grows, it usually pays to use incomplete

blocks, because error variance tends to grow with block size due to increasing within-block heterogeneity (John & Williams, 1995). Also, blocking can be done in two dimensions, giving rise to row–column designs (Piepho et al., 2015).

When using designs with one-way or two-way blocking, there is commonly a concern when randomization returns a design with replications of some treatments clustering in a certain part of the field. A common reaction in practice then is to discard the current design and generate a new one in the hope of obtaining a more even distribution of treatment replications across the field. We will henceforth abbreviate this feature as ED for 'evenness of distribution' (Piepho et al., 2016). Another desirable feature of a randomized design is neighbour balance, abbreviated here as NB, meaning that each treatment has varying neighbours across replications and the number of direct adjacencies is balanced between pairs of treatments. With common blocked and row–column designs, randomization may

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produce layouts with considerable neighbour imbalance. Piepho et al. (2018) considered two different approaches to achieve good properties in terms of both NB and ED. One is based on the imposition of blocking units in addition to rows and columns that ensure better ED, and certain spatial covariance structures that help achieving NB. The interplay of the various blocking units and spatial covariance is tuned by the choice of values for the variance of the associated random block effects and residual error. The other approach is model-free and avoids undesirable configurations around each plot by imposing restrictions during the computer search for design optimization. The model-based approach to design optimization can be implemented with packages such as OD-V2 (Butler & Cullis, 2018; Cullis et al., 2020) and the OPTEx procedure in the QC (quality control) module of SAS (Piepho, 2015). Some care is required regarding the appropriate choice of variance values. The model-free approach does not require such choices. Instead, it chooses an algorithm and a set of constraints on permissible spatial configurations during design search. One such choice (Piepho et al., 2018) has recently been implemented in the design package CycDesign 7.0 (<https://www.vsn.co.uk/software/cycdesign/>), which we will use here.

The purpose of this paper is to illustrate the flexibility of the model-free approach in the generation of row-column designs with good NB and ED (NB&ED) properties. There is nothing that is in itself new; we draw on previously published work that appeared in statistical journals. Our main intention here is to provide a broadly accessible tutorial that hopefully is appealing to agronomists and crop scientists. Five examples will be utilized to demonstrate the capabilities of the approach.

2 | MATERIALS AND METHODS

The objective criterion for the design search is the average efficiency factor, accounting for row and column blocks, which will be denoted here as E . The average efficiency factor E takes values between 0 and 1. If all treatments are equally replicated, we will use the g series definition of E , given by $2/(\bar{r}_a apv)$, where $\bar{r}_a$ is the arithmetic average of the numbers of treatment replications and $apv = 2[v(v-1)]^{-1} \sum_{i < j} (V_{ii} + V_{jj} - 2V_{ij})$, where v is the number of treatments and $V = \{V_{ij}\}$ is the Moore-Penrose inverse of the treatment information matrix (Williams & Piepho, 2015). It can be viewed as the average variance of a difference in treatment means for a completely randomized design, divided by the variance of a treatment difference in the current design, assuming that both designs have the same error variance. In practice, it may be expected that the use of incomplete blocks will reduce the plot error variance, thus ensuring that the actual variance of a difference in treatment means will be smaller for the blocked design. For designs with unequal treatment replication, we will use the e series definition of E based on canonical efficiency factors (John & Williams, 1995), given by $2/(\bar{r}_a apv_w)$, where $apv_w = 2[v(v-1)]^{-1} \sum_{i < j} r_i r_j \bar{r}_a^{-2} (V_{ii} + V_{jj} - 2V_{ij})$ and r_i is the number of replications for the i -th treatment (Williams

& Piepho, 2015). For equally replicated designs, the e and g series definitions coincide.

In the spatial optimization for a row-column design, several criteria are monitored in addition to E ; these assess the properties regarding ED and NB. One such criterion, targeting ED, is the minimum span in both the row and column directions. For example, a span of 7 for a treatment in column direction means that the leftmost and rightmost replications of the treatment are seven columns apart. Another criterion is the NB score. If there are n row neighbour occurrences of a pair of treatments, this will contribute $n(n-1)/2$ to the score. For instance, if a pair of treatments does not appear as row neighbours the score is 0, and if the pair appears three times the score is 3. The NB score is then obtained by summing over all treatment pairs.

We assume that plots are usually long and thin and so for non-resolvable designs, if the number of columns is greater than the number of rows, the NB score is just calculated on neighbours along rows, that is between adjacent long thin boundaries. If the number of rows is greater than the number of columns, the NB score is calculated between rows in the same column. For resolvable designs that are only latinized in the column direction, the NB score is calculated between rows in the same column. For other resolvable designs such as non-latinized row-column designs and designs latinized by rows and columns, the NB score is calculated in both directions.

A detailed description of the model-free approach to the construction of NB&ED designs is included in Piepho et al. (2018). Here, we will present an overview of the methods adopted in the construction of the examples in this paper. In Example 1, the starting point is an efficient non-resolvable row-column design, that is a design with optimal or near-optimal average efficiency factor. The columns are divided into columngroups (a feature also known as 't-latinization'; John & Williams, 2002), and treatments are swapped between columns such that no treatment is repeated in a columngroup. Next, the NB score is addressed and further treatment interchanges are conducted within columngroups to reduce this score. Then, spatial neighbours are identified and eliminated, if possible, with further within-columngroup treatment interchanges. Note that the term 'spatial neighbours' refers to different types of adjacencies of treatment replications such as diagonals, replications that are in adjacent columns or are two columns apart. There are a number of types of spatial neighbours, and they are treated in order of priority (Piepho et al., 2018). Because the columngroups do not have treatments repeated, it is only necessary to address the issue of spatial neighbours at the edges of the columngroups. In addition to these criteria to limit the proximities of treatment replications, the algorithm also tries to maximize the overall span of treatment replications as discussed in the previous paragraph.

A spatial objective function is constructed to incorporate all of the NB and ED considerations. The aim is to minimize this function, and as there are usually many designs obtaining this minimum, the algorithm is run many times to additionally try and increase the value of E . A feature of the approach is that for larger arrays, any drop in E

TABLE 1 Summary of dimensions and average efficiency factors (*E*) for designs generated for Examples 1–5 with and without optimization for neighbour balance and evenness of distribution (NB&ED) (Piepho et al., 2016, 2018)

Example	Design Type	Rows	Columns	Treatments (+standards)	<i>E</i> no NB&ED	<i>E</i> NB&ED
1	Non-resolvable row-column	9	13	34	0.850	0.821
2	Resolvable row-column	6	10	20	0.342	0.332
3	Single-location p-rep	25	12	252 (+12)	0.524	0.411
4	Multi-location p-rep	18 ^a	30 ^a	450	0.912	0.910
5	Multi-location p-rep	15 ^a	28 ^a	330 (+24)	0.891	0.888

^aThese are numbers per location.

due to the introduction of NB and ED constraints is likely to be small. In any case, the algorithm has an inbuilt protection to ensure that there is not a large drop in *E* as a result of achieving a minimum value for the spatial objective function. Whilst it is acknowledged that the construction of experimental designs is traditionally focussed on maximizing the average efficiency factor, the spatial design properties are also an important consideration, especially if some type of spatial analysis is envisaged (Hoefer et al., 2020; Mao et al., 2020). These days readily available computers and software have greatly increased this likelihood.

The remaining examples introduce variations on the general approach outlined above. Example 2 is a resolvable design (i.e. a design having complete replicates) and hence there is no strict need to construct the initial column groups, though we do use such groups later during the design generation process in order to improve ED. Examples 3–5 relate to partially replicated (p-rep) designs (Cullis et al., 2006). In these designs, a subset of the treatments is duplicated, whereas the remaining treatments only have a single replication. The duplicated treatments serve to provide an estimate of error and also to allow for block adjustments (Williams et al., 2014). P-rep designs can be compared with augmented designs, where all treatments are unreplicated, except for a few standard treatments (Federer et al., 1975). An advantage of p-rep designs is that all plots are devoted to the treatments of interest, whereas augmented designs devote a substantial number of plots to standards for assessment of error and block effects. Often, standards are of interest in themselves and so p-rep designs may be extended to comprise a few plots for standards, as will be demonstrated in Examples 3 and 5.

Example 3 is a single-location p-rep design where the number of duplicated test treatments and standard treatments can be specified. If standard treatments are included, they are first arranged such that the number in any row or column does not differ by more than one; spatial neighbours are also reduced as much as possible. The duplicated test treatments are then considered, and interchanges between all test treatments are carried out to improve the row and column spans of the duplicates. As mentioned above, the algorithm is run many times to improve the average efficiency factor which is calculated to include both standards and test treatments.

Examples 4 and 5 are multi-location p-rep designs where the total number of replications of the test treatments across locations

is the same for all treatments. The user can select the number of test treatments and locations and the number of replications of each test treatment. There is then a menu of choices available for the combination of the number of standards (it can be zero), duplicated test treatments and the (common) row and column dimensions of the field layout at each location. The algorithm searches for an efficient experimental design to fit these specifications using the method described by Williams et al. (2014) based on alpha arrays. This construction method results in duplicated treatments having one replication in the first half of the columns and the other replication in the second half. In addition, at each location the spread of standards (if specified) and row and column spans of the duplicated test treatments are addressed as discussed for single-location p-rep designs. Finally, the algorithm is run repeatedly to improve the average efficiency factor which is calculated with respect to rows and columns within locations and includes both standards and test treatments.

We evaluate the efficiency of a design based on a linear model with fixed effects for treatments, rows and columns, and this is also the baseline model we suggest for analysis, with an option to take row and column effects as random for recovery of inter-block information. When the design is resolvable, the baseline model also includes a fixed effect for replicates (Example 2). In case of multi-location p-rep designs (Examples 4 and 5), the row and column effects are nested within locations, and for analysis, the model also comprises effects for locations and treatment–location interaction. The latter two effects are random when locations are considered as a random sample from a larger target population of environments, and they would be modelled as fixed otherwise. These baseline models can be extended to cover specific features of the experiment, such as inclusion of terms for covariates and add-on components for spatial covariance of plot errors (Piepho & Williams, 2010).

3 | RESULTS

Table 1 summarizes the results for the five examples. In each of the examples, there is a single treatment factor. In order to emphasize context, we will use subject-specific denominations of the treatment factor, which are introduced at the beginning of each example.

3.1 | Example 1

In a regional post-registration yield trial conducted by the Landesforschungsanstalt für Landwirtschaft und Fischerei Mecklenburg-Vorpommern, Gültow, Germany, 34 varieties of silage maize were to be tested in a non-resolvable layout with 9 rows and 13 columns. The treatment factor is variety. The dimensions of the experiment meant having either three or four replications per variety. Specifically, the first 19 varieties have three replications and the last 15 varieties have four replications. Note that normally variety numbers would be randomized to the unequally replicated varieties, but for convenience we have not done this. In these trials, there is concern with randomizations where replications of a variety are clustered in one area of the field, when a row-column design is used. An example of poor ED is given in Figure 1a, where, for example, three replications of variety 21 are clustered closely together in the middle columns of the layout. That same variety, as well as varieties 10, 15, 25, 29 and 31, also shows diagonal self-adjacencies. Such a randomization could adversely affect the mean performance estimate of those varieties when productivity in the relevant area of the layout is markedly reduced due to an environmental or management constraint, for example disease or low soil fertility. Another concern is the repeated occurrence in the same neighbourhood of the same pair of varieties. For example, varieties 32 and 8 have adjacencies in rows 3 and 8, which constitutes an instance of poor NB. It should

(a)

24	18	3	25	9	32	20	15	19	17	26	2	34
34	27	16	14	25	6	15	21	11	24	30	12	9
30	20	2	28	13	21	7	4	14	34	33	32	8
5	11	17	33	21	29	25	34	13	23	3	27	26
7	29	30	24	28	26	23	31	10	4	15	22	11
29	32	31	20	22	9	5	27	30	10	8	17	1
19	26	21	1	10	33	16	22	6	25	20	7	2
33	16	24	27	18	31	13	19	32	8	12	23	28
18	4	28	29	12	23	31	1	3	6	22	14	5

(b)

25	32	14	22	29	4	19	21	26	11	24	17	28
21	23	8	1	24	33	25	30	31	22	10	27	5
7	29	2	31	21	27	12	20	16	9	33	15	34
24	20	11	30	28	34	14	6	18	3	13	32	26
28	9	27	6	25	26	13	17	2	7	19	12	22
31	4	33	16	18	11	10	28	15	8	14	21	1
30	3	5	19	20	32	1	33	27	34	29	31	23
16	34	17	12	23	7	22	24	5	25	4	20	8
26	18	13	10	15	9	3	29	32	23	30	6	2

FIGURE 1 Row-column design for $v = 34$ varieties, $k = 9$ rows and $s = 13$ columns (Example 1). (a) Design with row and column randomization but no further restrictions on randomization. Examples of plots displaying poor ED or NB are boldfaced. Minimum span 4 in columns and 3 in rows; NB score = 1; average efficiency factor $E = 0.850$. (b) Design with NB&ED restrictions on randomization. Minimum span 8 in columns and 6 in rows; NB score = 0; average efficiency factor $E = 0.821$. Column groups are highlighted by alternating white and grey background

be pointed out here that we assume plots to be longish down the column, so adjacencies within the same row are of primary importance. A particularly drastic scenario is where one of the two varieties suffers from severe environmental stress such as frost damage and dies, so that the varieties on the neighbouring plots are at an advantage in their competition for light and other resources. This often prompts experimenters to also discard the two neighbouring plots in the same row. If the same neighbourhoods with a particular variety occur repeatedly, that variety can also suffer from a substantial loss of replication, even though it was not itself affected by frost damage. In fact, it was this kind of scenario that initially prompted the development of alternative design approaches in Piepho et al. (2016), Piepho et al. (2018), including the NB&ED feature considered here. Just which exact scenario will materialize in a given trial is unpredictable. Ensuring both NB and ED can be viewed as an insurance policy against any such unforeseen events. The design generated for this example using the NB&ED option (called 'spatial' in CycDesignN) is shown in Figure 1b. There is no pair of varieties that has more than one direct adjacency. The minimum spans for varieties are 8 and 6 for columns and rows, respectively. Also, all varieties have their replications well spread across the layout. The premium to pay for this improved NB&ED is a slight drop in the average efficiency factor from $E = 0.850$ in Figure 1a to $E = 0.821$ in Figure 1b. This slight drop may be considered to be more than offset by the improved NB&ED achieved by the restriction on randomization.

This example may be used to illustrate how the model-free approach works, which is described in more detail in Piepho et al. (2018) under the divisible method. To improve on ED, the columns are first divided into a number of column groups. The aim is make sure that no treatment occurs more than once in any column group. The column groups are determined as follows. Let v , k and s denote the numbers of treatments, rows and columns of the design, respectively. Further, let t be the smallest integer value such that $k(t-1) > v$ and m and q the integer and fractional parts of s/t . Then, the algorithm forms m column groups of size t (hence the term t -latinization) and one column group of size $p = t \times q$. In our example, we have $v = 34$, $k = 9$ and $s = 13$, such that $t = 3$, $m = 4$, $q = 1/3$ and $p = 3 \times 1/3 = 1$. Thus, starting from the left in Figure 1b, we form $m = 4$ column groups with $t = 3$ columns each and one column group with a single column ($p = 1$; at the right end of Figure 1b). It is readily verified that none of the treatments appears more than once in any of these column groups as required.

3.2 | Example 2

A rectangular grassland area was to be sampled on $v = 20$ dates in a project conducted by the State Institute of Agricultural Engineering and Bioenergy, University of Hohenheim. The treatments in this case correspond to the dates. The area was subdivided into 60 plots, arranged on a 6×10 rectangular grid. The six rows were divided into three blocks, each composed of two rows. The original plan was to allocate dates to plots within blocks

according to a randomized complete block design. Inspecting the layout for this original plan (Figure 2a), it was realized that the sampling dates were not very well distributed between the columns. For example, the three replications for date 9 all appeared in the second column. Date 18 also displays poor ED with respect to the columns it appears in. Moreover, the average efficiency factor is quite low ($E = 0.162$). Furthermore, the date pair (1, 7) appears next to each other in two rows (1 and 4), which is undesirable in terms of NB. In order to avoid these problems, an alternative design was sought. A standard approach is to use a resolvable row-column design (John & Williams, 1995; Piepho et al., 2015), in which complete replicates are arranged in two rows and ten columns. In this design, a restriction can be imposed that no dates will appear more than once down a long column of six plots (the latinized property). This restriction can be further extended to groups of t contiguous columns (the t -latinized property). Here, we use $t = 3$ because a group of three columns will comprise 18 plots, which is almost the total number of dates ($v = 20$). As there are ten columns altogether, the rightmost column will be considered as a fourth column group. This means that the minimum column span for $t = 3$ should be at least 5. In addition, we can invoke the NB&ED option to ensure ED of replications across the area. Such a design is shown in Figure 2b. With this example, there is particular value in the NB&ED construction at the edges of the contiguous

replicates in terms of separating the same date from adjacent replicates. This is especially the case with the example having only two rows per replicate. Also note that this design loses very little in terms of efficiency for the NB&ED gains ($E = 0.342$ for an efficient latinized row-column design down to $E = 0.332$).

Most researchers would probably have rejected the randomized block design in Figure 2a outright and tried to improve it, for example by some ad hoc swaps of dates. The statistical properties of the resulting design would not, however, be obvious. The advantage of the t -latinized row-column design in Figure 2b is that it is obtained by a defined algorithm with a clearly specified two-way blocking structure that in combination with the constrained search ensures good ED properties whilst maintaining an average efficiency factor close to a design without the NB&ED feature.

3.3 | Example 3

Clarke and Stefanova (2011) consider a single-location p-rep design augmented with standards. It has $v = 252$ lines, of which 36 are duplicated. In addition, there are 12 standard plots. The treatments are the lines and standard varieties. The layout has $k = 25$ rows and $s = 12$ columns. Figure 3 shows a design with these specifications, and as with the Clarke and Stefanova design, there is no more than one standard in any row or column.

The average efficiency factor is $E = 0.411$ for this design. The minimum span is 11 for rows and 6 for columns. By comparison, the design in Figure 2f of Clarke and Stefanova (2011) has $E = 0.270$, and the minimum spans are down to 1 for several lines in the column direction (219, 223, 227, 234) and 251 in the row direction.

3.4 | Example 4

The State Breeding Institute of the University of Hohenheim conducts trials at multiple locations with partial replication at each location. Consider a trial with 450 breeding lines of triticale at 10 locations. The treatments here are the lines. Typically, 20 per cent of the lines are duplicated at each location in order to allow for an estimation of the error variance and for block adjustments. This means that for each line, there are 12 replications across the 10 locations. The first location of a multi-location row-column p-rep design (Williams et al., 2014) generated with NB&ED optimization is shown in Figure 4. The full design is available with the Appendix S1 and S2 (file example4.html). All locations have a minimum span of 15 for columns and 9 for rows. The average efficiency factor is $E = 0.910$.

3.5 | Example 5

Nordsaat Saatzucht GmbH routinely conducts p-rep trials with winter barley at five locations. Assume that 330 lines are tested, and

(a)

13	9	19	10	3	11	17	1	7	6
20	8	12	2	18	4	14	5	16	15
5	17	3	14	12	4	15	8	11	13
16	9	10	18	7	1	6	20	19	2
12	15	7	2	19	6	5	20	1	11
4	9	8	18	10	3	16	13	17	14

(b)

9	19	15	2	8	11	1	4	16	3
12	7	18	10	6	5	14	17	20	13
16	11	13	14	15	7	9	12	18	2
10	20	1	3	17	4	19	6	8	5
8	14	2	19	20	12	15	13	7	17
3	4	6	18	9	1	5	11	10	16

FIGURE 2 Two designs for Example 2. (a) Original sampling plan laid out as a randomized complete block design for $v = 20$ dates and $r = 3$ replicates with replicates formed by two adjacent rows. The replicates are delineated by the two thick lines. Minimum span 1 in columns and 4 in rows; NB score = 4; average efficiency factor $E = 0.162$. (b) Resolvable row-column design for $v = 20$ dates, $r = 3$ replicates, $k = 2$ rows within replicates and $s = 10$ columns within replicates, t -latinized ($t = 3$) down the columns and with NB&ED feature. The replicates are delineated by the two thick lines. Minimum span 5 in columns and 4 in rows; NB score = 0; average efficiency factor $E = 0.332$. Column groups are highlighted by alternating white and grey background

75	183	69	134	117	52	140	12	S1	190	131	111
57	172	S1	60	76	240	20	27	59	171	70	95
15	198	78	209	196	231	26	1	152	220	18	108
210	188	124	113	153	186	176	86	51	215	53	37
64	138	42	247	32	165	125	238	80	97	218	S1
81	159	87	158	S1	230	234	94	233	62	118	245
106	167	216	243	7	219	72	228	43	239	136	249
161	100	229	31	89	93	73	143	16	24	199	109
145	66	114	137	201	132	217	178	236	241	45	246
105	101	168	46	2	34	29	157	160	56	67	181
13	156	14	200	128	83	S1	189	149	19	9	126
36	148	85	131	6	99	248	65	212	211	221	141
S1	8	11	163	139	226	68	154	58	44	84	252
194	192	164	185	3	147	57	4	175	90	91	28
123	26	179	41	162	S1	155	23	195	122	74	197
112	25	225	55	62	150	151	115	235	S1	117	10
47	88	251	220	71	129	187	116	206	159	243	33
121	49	135	249	170	35	222	40	107	38	103	247
130	S1	143	119	202	213	191	244	17	188	208	63
193	140	59	16	92	207	224	77	198	203	96	167
180	102	45	67	239	82	54	166	5	232	S1	39
215	142	223	110	221	174	169	98	184	172	79	61
127	29	227	S1	177	30	182	242	205	146	120	78
58	37	204	95	104	15	194	237	173	48	231	93
133	21	1	144	214	250	109	S1	50	114	22	124

FIGURE 3 A single-location partially replicated (p-rep) design for $v = 252$ lines, of which 36 are duplicated (printed in boldfaced black) and 216 are unreplicated (printed in blue), and 12 standard plots (printed in boldfaced red) (Example 3). The layout has $k = 25$ rows and $s = 12$ columns. Minimum span for duplicate treatments: 11 for rows and 6 for columns; average efficiency factor $E = 0.411$

at each location, 20% of the lines, that is 66 lines, are duplicated. An additional 24 plots are to be devoted to standard varieties. The treatments in this case are the lines and standard varieties. The layout has 15 rows and 28 columns. Figure 5 shows a design for one location. It has an average efficiency factor of $E = 0.888$. The minimum span for test lines is 8 for rows and 14 for columns at all locations. There is a single symbol 'S' for the standard plots, but several different standard varieties could be used.

Previously, the company used augmented designs with no duplication of lines, but plenty of replications for standards. Thus, in augmented designs used in the past, the proportion of plots devoted to standards was larger than for the design in Figure 5. It may be argued that with partial replication of lines taking over the role of standards, the number of plots devoted to standards can be somewhat reduced. A primary reason for this is that duplicated lines can be used in conjunction with replicated standards to contribute to reliable error estimation and block adjustment.

4 | DISCUSSION

The design approach we used here is focussed on two-dimensional layouts on a rectangular grid of experimental units. The five examples were from field trials, but the same kinds of layout can be used in greenhouses and in-house phenotyping platforms. The main blocking factors are rows and columns, and in case of non-resolvable designs, these are the only ones. In the case of resolvable designs, there are replicates, and the rows and columns may also be latinized along rows and columns, giving rise to additional blocking factors. NB and ED are ensured by imposing spatial restrictions during the design search.

From a researcher's point of view, the average efficiency factor, which we use for design optimization, needs to be balanced against other properties of the design, and ED and NB may be given preference. This would normally mean that some loss in average efficiency factor can be tolerated if NB&ED properties can

169	44	155	148	286	402	217	175	442	384	422	303	197	389	278	37	246	185	352	62	98	341	289	262	139	265	328	247	162	160
137	400	405	260	324	179	210	287	330	193	176	216	45	392	40	230	441	111	135	339	350	121	448	367	420	15	16	215	196	353
91	159	432	437	266	152	194	107	76	404	310	239	333	205	391	168	222	115	450	360	86	443	19	257	419	82	438	329	326	38
428	254	118	344	75	449	71	255	83	70	269	406	228	370	343	355	2	77	9	72	218	253	173	171	104	112	429	140	358	377
142	116	59	362	379	172	92	416	271	130	150	132	229	354	119	248	144	23	279	444	407	412	264	146	282	212	237	270	100	1
251	312	156	65	48	190	211	241	337	149	387	95	293	90	338	375	68	376	7	42	50	55	203	145	22	242	103	421	93	122
294	403	120	219	446	365	447	29	138	371	32	183	263	347	272	348	232	250	204	301	114	24	67	147	182	414	258	220	288	390
170	309	17	26	307	238	296	178	165	275	357	252	235	401	399	418	245	425	108	274	20	11	368	433	131	109	346	46	427	283
5	415	361	106	41	94	342	325	397	78	36	198	359	221	14	99	434	158	54	280	34	321	302	47	154	233	320	134	177	234
16	157	58	185	352	297	281	121	382	285	292	186	328	336	345	400	28	141	214	402	69	124	148	304	384	380	3	323	210	356
226	350	443	6	438	381	60	4	128	267	276	409	445	66	174	208	308	97	317	424	51	107	76	322	423	163	195	435	298	89
143	377	98	383	326	436	35	244	386	249	394	273	314	12	223	61	13	79	75	306	378	194	430	266	269	303	56	96	10	136
171	82	144	393	385	73	259	412	19	43	38	212	237	270	153	254	74	362	315	172	25	416	231	87	150	374	155	428	53	119
30	375	199	102	236	187	50	105	203	117	22	57	421	261	340	63	405	369	151	21	71	327	33	191	176	260	228	156	389	113
114	429	319	358	364	133	318	291	18	373	164	184	213	166	439	31	126	417	80	365	447	300	431	206	371	295	299	92	338	85
332	363	182	313	108	283	256	188	240	433	351	225	49	167	426	44	334	268	324	238	296	64	330	309	180	413	84	219	391	410
264	209	434	207	390	189	277	52	302	147	265	233	202	316	100	129	311	227	41	396	224	305	192	395	357	132	201	27	399	243
141	366	245	125	81	62	101	321	411	349	110	372	200	284	408	331	127	388	290	440	161	335	123	181	398	186	88	8	39	336

FIGURE 4 A partially replicated (p-rep) design for $v = 450$ lines, 10 locations and 12 replications per line (Example 4). At each location, the design has $k = 18$ rows and $s = 30$ columns. The layout shown is for the first location. It has 90 duplicated lines, printed in boldfaced black. The unreplicated lines are printed in blue. Minimum span for duplicate treatments: 9 for rows and 15 for columns at all locations; average efficiency factor $E = 0.910$

109	101	127	17	189	311	223	7	55	118	86	37	329	83	214	253	105	261	270	30	72	249	10	143	S	100	296	297
272	27	283	48	159	315	132	226	38	164	44	S	183	266	202	322	6	191	250	251	201	95	273	259	302	179	220	21
84	293	13	140	285	115	319	82	74	218	146	290	216	S	20	317	310	34	122	208	282	90	274	57	26	193	75	213
22	58	124	S	263	294	291	219	286	197	121	71	211	80	262	150	S	156	307	243	328	1	225	41	61	300	5	295
112	73	212	188	277	241	185	23	S	155	62	163	172	142	106	230	129	267	S	221	239	94	78	173	91	108	158	194
168	141	308	133	245	14	S	138	136	66	97	88	278	255	169	154	45	196	148	235	228	276	284	126	49	167	S	195
236	210	12	9	S	181	125	113	292	29	76	145	64	279	186	246	24	232	120	S	170	151	254	314	134	98	114	8
105	214	117	280	65	270	175	72	238	224	161	147	229	222	176	205	18	S	42	131	47	198	160	S	269	231	190	311
33	68	287	4	63	313	323	16	275	271	S	242	21	184	92	104	107	27	103	248	S	96	118	264	298	199	53	25
85	57	59	110	90	178	102	95	179	215	312	26	S	174	304	70	204	207	165	84	315	325	31	223	320	115	281	S
54	128	187	77	137	S	328	247	220	203	152	61	300	99	306	13	58	123	180	3	40	S	286	35	157	257	7	71
87	153	225	217	267	139	111	S	51	244	256	330	2	305	182	S	39	69	188	285	32	268	155	288	81	172	15	227
200	43	S	284	93	192	318	209	289	89	75	116	167	303	67	124	321	19	308	130	189	237	177	79	66	185	218	97
1	201	24	30	56	299	78	265	28	52	173	206	162	144	240	50	9	236	181	210	258	292	46	309	73	S	252	301
307	S	253	327	3	42	171	36	166	231	269	326	234	60	119	324	135	65	11	260	113	140	316	233	149	12	145	245

FIGURE 5 A partially replicated (p-rep) design for $v = 330$ lines, 5 locations and 6 replications per line (Example 5). At each location, the design has $k = 15$ rows and $s = 28$ columns. The layout in this figure is for one location. It has 66 duplicated lines, printed in boldfaced black, and 24 standard plots, printed in boldfaced red. The unreplicated lines are printed in blue. Minimum span for duplicate treatments: 8 for rows and 14 for columns at all locations; average efficiency factor $E = 0.888$

be improved. The strategy taken here is to impose restrictions in the design search that directly target both NB and ED properties and within those constraints to optimize the average efficiency factor. Our objective is to not achieve extreme examples of ED, such as the knights move Latin squares that Tedin (1931) studied, as this could result in increased error variance bias (Williams & Piepho, 2019). Thus, we mainly aim to provide a reasonable guard against clumping, without wanting to impose a very specific pattern.

Our model-free approach also involves the formation of column-groups but we do not propose to include such groups in a baseline model, because effects for columns are included in that model and would largely absorb any effects associated with the column-groups. Also, since the column-groups are not orthogonal to treatments (i.e. the design is not resolvable), their inclusion would not provide the benefit of being able to remove variation containing no treatment information. In addition, because the grouping is arbitrary it is unlikely that the variance components for the column-groups will be substantially greater than those for the columns (not like is often the case when we remove replicate variation). The spatial restrictions imposed on neighbourhood configurations imposed under the model-free approach imply restrictions on randomization. We have investigated the error variance bias due to the randomization restriction and found this to be acceptably small based on analysis of uniformity trials (Williams & Piepho, 2019).

Despite the various spatial constraints imposed in our design search, our sole optimality criterion is the average efficiency factor E assuming a baseline model with fixed effects for rows and columns. This criterion bears some direct relationship with other measures often used in breeding applications, such as heritability and response to selection (Isik et al., 2017). It may be asked how the spatial constraints imposed during the search could be accounted for in an assessment of efficiency, also compared with more classical row-column designs. The answer is that the spatial constraints are so many and, perhaps more importantly, that none of these seem amenable to direct representation by an additional model component, a consequence of the model-free, purely algorithmic nature of the whole approach.

We have considered t-latinization with the resolvable design in Example 2. Whereas the model-free approach automatically determines the value of t with non-resolvable designs as illustrated with Example 1, in generating a resolvable design, it is necessary to actively specify the value of t . A good strategy is usually to choose the value so that the product $k \times t$ is close to, but not larger than the number of treatments v . In Example 2, there are 4 column-groups of sizes 3, 3, 3 and 1. Such latinization is of particular value when rows or columns have many plots. A very common design in practice has a complete replicate in each row but incomplete blocks down the column. Such a row-column design is non-resolvable by definition because it only has a single row per replicate. When the number of

columns is large, with $v = 39$ treatments say, and the number of rows is small, $k = r = 4$ say, then latinization will ensure good ED properties of the design.

Occasionally, when the number of columns per columngroup does not divide v , then despite latinization, ED for an individual treatment may appear less than optimal. For example, consider the distribution of the three replications of treatment 5 in Figure 2b. The left part of the field does not have a replication of that treatment, and the 'barycenter' (average of the row and column coordinates) of the three plot positions is located in the right half. Super-imposing three non-rectangular blocks of equal size would shift the barycenter somewhat towards the middle of the field, and the three replications would show even better coverage of the whole field.

Cullis et al. (2006) noted that for single-location p-rep designs 'the optimal allocation of duplicated and control lines [...] is still an open question', and this point is re-iterated in Clarke and Stefanova (2011). The construction approach discussed here, as demonstrated in Example 3, now presents a flexible way forward. On the other hand for multi-location p-rep designs, CycDesign adopts the procedure mapped out by Williams et al. (2014) and as such requires equal replication of test treatments when considered across all locations. In practice, this can sometimes be limiting. One possibility to accommodate for unequal replication could be to construct a series of single-location p-rep designs and coordinate these across locations. This could be a topic for further software development.

In Example 1, we mentioned that researchers occasionally discard plots next to a plot lost to frost damage or other environmental stresses because such plots are bound to yield-biased responses, particularly when plots are long and thin down the columns so that within-row neighbour effects are likely. From a statistical point of view, it is desirable to not discard valuable data. If data from such plots are kept, however, adjustments must be made in the statistical analysis to correct for neighbour effects, and such adjustments may also adversely affect the precision of the design. Optimizing the NB of a design will also help in this scenario in that it will balance out any efficiency losses due to such adjustments between treatment pairs.

On a final note, we would like to emphasize that different runs of the search algorithm can yield slightly different average efficiency factors and that it may therefore be worth running the algorithm several times. For example, in Example 2, we were able to occasionally get an average efficiency factor of $E = 0.339$ for the design with NB&ED adjustment after several runs, a slight improvement over the design reported in Figure 2b, where $E = 0.332$.

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DATA AVAILABILITY STATEMENT

No data is needed for this paper.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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